

Experiment 6: Functions

Experiment 6.1

```
C exp6_1.c > ...
1 // Develop a recursive and non-recursive function FACT(num) to find the factorial of
2 // a number, n!, defined by FACT(n) = 1, if n = 0. Otherwise, FACT(n) = n * FACT(n-
3 // 1). Using this function, write a C program to compute the binomial coefficient.
4 // Tabulate the results for different values of n and r with suitable messages.
5 #include <stdio.h>
6
7 // Recursive function to calculate factorial
8 int FACT_recursive(int n) {
9     if (n == 0)
10         return 1;
11     else
12         return n * FACT_recursive(n - 1);
13 }
14
15 // Non-recursive (iterative) function to calculate factorial
16 int FACT_nonrecursive(int n) {
17     int fact = 1;
18     for (int i = 1; i <= n; i++) {
19         fact *= i;
20     }
21     return fact;
22 }
23
24 // Function to compute binomial coefficient using factorial
25 int binomial_coefficient(int n, int r, int use_recursive) {
26     int n_fact, r_fact, n_r_fact;
27
28     if (use_recursive) {
29         n_fact = FACT_recursive(n);
30         r_fact = FACT_recursive(r);
31         n_r_fact = FACT_recursive(n - r);
32     } else {
33         n_fact = FACT_nonrecursive(n);
34         r_fact = FACT_nonrecursive(r);
35         n_r_fact = FACT_nonrecursive(n - r);
36     }
37
38     return n_fact / (r_fact * n_r_fact);
39 }
```

```

41 int main() {
42     int n, r, choice;
43
44     printf("Choose factorial method:\n");
45     printf("1. Recursive\n");
46     printf("2. Non-Recursive\n");
47     printf("Enter choice (1 or 2): ");
48     scanf("%d", &choice);
49
50     printf("\nEnter value of n: ");
51     scanf("%d", &n);
52
53     printf("Enter value of r: ");
54     scanf("%d", &r);
55
56     if (r > n || n < 0 || r < 0) {
57         printf("\nInvalid input! Ensure that 0 <= r <= n.\n");
58         return 0;
59     }
60
61     int result = binomial_coefficient(n, r, choice == 1);
62
63     printf("\nUsing %s method:\n", choice == 1 ? "Recursive" : "Non-Recursive");
64     printf("Binomial Coefficient C(%d, %d) = %d\n\n", n, r, result);
65
66     // Table of binomial coefficients up to n
67     printf("Table of Binomial Coefficients (for n up to %d):\n", n);
68     for (int i = 0; i <= n; i++) {
69         for (int j = 0; j <= i; j++) {
70             printf("%d\t", binomial_coefficient(i, j, choice == 1));
71         }
72         printf("\n");
73     }
74
75     return 0;
76 }
77

```

Experiment 6.2

```

/ Develop a recursive function GCD (num1, num2) that accepts two integer arguments. Write a C program that invokes this function to find the greatest common divisor of two given
integers.
#include <stdio.h>

/ Recursive function to find GCD using Euclidean algorithm
int GCD(int num1, int num2) {
    if (num2 == 0)
        return num1;
    else
        return GCD(num2, num1 % num2);
}

int main() {
    int a, b, result;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    if (a < 0) a = -a; // handle negative inputs
    if (b < 0) b = -b;

    result = GCD(a, b);

    printf("The GCD of %d and %d is: %d\n", a, b, result);

    return 0;
}

```

Experiment 6.3

```
C exp6.3.c > ...
1 // Develop a recursive function FIBO (num) that accepts an integer argument. Write a C program that invokes this function to generate the fibonacci sequence up to num.
2 #include <stdio.h>
3
4 int FIBO(int num) {
5     if (num == 0)
6         return 0;
7     else if (num == 1)
8         return 1;
9     else
10        return FIBO(num - 1) + FIBO(num - 2);
11 }
12
13 int main() {
14     int n, i;
15
16     printf("Enter the number of terms: ");
17     scanf("%d", &n);
18
19     if (n <= 0) {
20         printf("Please enter a positive integer.\n");
21         return 0;
22     }
23
24     printf("Fibonacci sequence up to %d terms:\n", n);
25     for (i = 0; i < n; i++) {
26         printf("%d\t", FIBO(i));
27     }
28     printf("\n");
29
30     return 0;
31 }
32
```

Experiment 6.4

```
C exp6_4.c > ISPRIME(int)
1 // Develop a C function ISPRIME (num) that accepts an integer argument and
2 // returns 1 if the argument is prime, a 0 otherwise. Write a C program that invokes
3 // this function to generate prime numbers between the given ranges.
4 #include <stdio.h>
5
6 int ISPRIME(int num) {
7     if (num <= 1)
8         return 0;
9
10    for (int i = 2; i * i <= num; i++) {
11        if (num % i == 0)
12            return 0;
13    }
14    return 1;
15 }
16
17 int main() {
18     int lower, upper;
19
20     printf("Enter the lower and upper bounds: ");
21     scanf("%d %d", &lower, &upper);
22
23     if (lower > upper) {
24         printf("Invalid range! Lower bound should be <= upper bound.\n");
25         return 0;
26     }
27
28     printf("Prime numbers between %d and %d are:\n", lower, upper);
29
30     for (int num = lower; num <= upper; num++) {
31         if (ISPRIME(num))
32             printf("%d\t", num);
33     }
34
35     printf("\n");
36     return 0;
37 }
38
```